

SHWETHA S – PES1UG24CS452 – BPS#24

LAB 2: AGILE BACKLOG CREATION & SPRINT SIMULATION IN JIRA

Public Bus Live Tracking & Crowding Estimator

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Project Key: PBLTCE

Domain: Smart Cities, Transport & Logistics

Methodology: Agile / Scrum

Problem Statement: BPS#24 – Public Bus Live Tracking & Crowding Estimator

1. OBJECTIVE

To use Jira to convert functional requirements into Agile backlog items, organize them into Epics and User Stories, prioritize and estimate them using Story Points, simulate two Sprints, and analyze the progress using Jira's Burndown Chart.

2. PROJECT OVERVIEW

The project is a **Public Bus Live Tracking & Crowding Estimator** designed for a Smart Cities, Transport & Logistics scenario.

The system collects GPS data from buses, provides real-time bus tracking, estimates bus arrival times, determines crowding levels, and provides useful information to commuters and fleet controllers.

The project was implemented in Jira using the **Agile Scrum methodology**.

3. EPICS

The functional requirements were divided into five major Epics.

Epic	Epic Name	Description
Epic 1	Real-Time Bus Tracking	Enables tracking and displaying the live location of buses using GPS data.
Epic 2	ETA Estimation	Provides estimated arrival times for buses.
Epic 3	Crowding Estimation	Estimates and displays the crowding level of buses.
Epic 4	Commuter Information	Provides current bus information to commuters.
Epic 5	Fleet Monitoring	Helps fleet controllers monitor buses and identify GPS or crowding-related issues.

4. USER STORIES

The Epics were divided into User Stories following the Agile format:

As a [user], I want [goal], so that [benefit].

The backlog contained **15 User Stories** distributed across the five Epics. **Epic**

1 – Real-Time Bus Tracking

- **Story 1.1:** Ingest GPS coordinates from buses
- **Story 1.2:** View real-time bus location
- **Story 1.3:** Monitor GPS data from buses

Epic 2 – ETA Estimation

- **Story 2.1:** View estimated arrival time

- **Story 2.2:** Calculate bus arrival time
- **Story 2.3:** Receive updated ETA information

Epic 3 – Crowding Estimation

- **Story 3.1:** Calculate crowding level of buses
- **Story 3.2:** View updated crowding information
- **Story 3.3:** Search for available buses **Epic**

4 – Commuter Information

- **Story 4.1:** View bus location, ETA, and crowding information
- **Story 4.2:** Display current bus information clearly
- **Story 4.3:** View status of all tracked buses **Epic**

5 – Fleet Monitoring

- **Story 5.1:** Identify buses with missing or delayed GPS data
- **Story 5.2:** Monitor crowding information across buses
- **Story 5.3:** [Use the exact Story 5.3 title shown in your Jira backlog]

Important: Before submitting, replace the Story 5.3 placeholder with the exact title visible in your Jira project.

5. BACKLOG PRIORITIZATION

The User Stories were prioritized according to their importance, user impact, and business value.

The Jira backlog used the priority levels:

- **High** • **Medium** • **Low**

The stories were reordered according to their priority before Sprint Planning.

6. STORY POINT ESTIMATION

Story Points were assigned using the **Fibonacci sequence** as required by the lab.

The Fibonacci scale used for estimation includes:

0, 1, 2, 3, 5, 8, 13, 20, 40, 100

Story Points represent the relative effort, complexity, risk, and uncertainty involved in completing a User Story.

The 15 User Stories were estimated and distributed across two Sprints. **Sprint Summary**

Sprint	Number of Stories	Total Story Points
Sprint 1	7	34
Sprint 2	8	35
Total	15	69

7. SPRINT 1 Sprint Goal

To implement the initial real-time bus tracking and ETA-related functionality.

Sprint 1

- Duration: **1 week**
- Number of User Stories: **7**
- Total Story Points: **34**

The selected stories were moved through the Scrum workflow:

To Do → In Progress → Done

All seven User Stories were successfully completed and Sprint 1 was completed in Jira.

Screenshot 1 Jira Backlog with Epics, User Stories, Priorities and Story Points

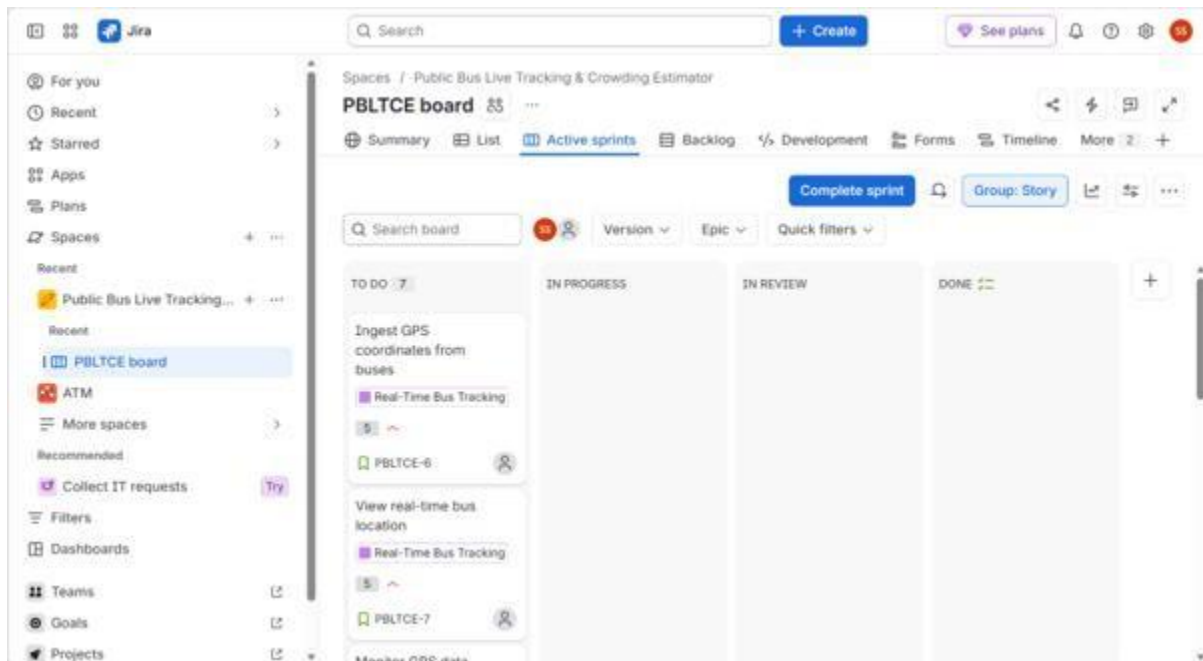
This screenshot shows the Jira Backlog for the 'PBLTCE board' under the space 'Public Bus Live Tracking & Crowding Estimator'. The backlog contains 15 work items. The first item is a 'Create a whiteboard to plan your work' task. The following items are user stories, each with a key, description, priority, status, story points, and assignee.

Key	Description	Priority	Status	Story Points	Assignee
PBLTCE-6	Ingest GPS coordinates from buses	Real-Time...	To Do	5	[User]
PBLTCE-7	View real-time bus location	Real-Time...	To Do	5	[User]
PBLTCE-8	Monitor GPS data from buses	Real-Time...	To Do	3	[User]
PBLTCE-9	View estimated arrival time	ETA Estim...	To Do	5	[User]
PBLTCE-10	Calculate bus arrival time	ETA Estim...	To Do	8	[User]
PBLTCE-11	Receive updated ETA information	ETA Estim...	To Do	3	[User]
PBLTCE-12	View crowding level of a bus	Crowding ...	To Do	5	[User]
PBLTCE-13	Calculate crowding level of buses	Crowding ...	To Do	8	[User]
PBLTCE-14	View updated crowd...				

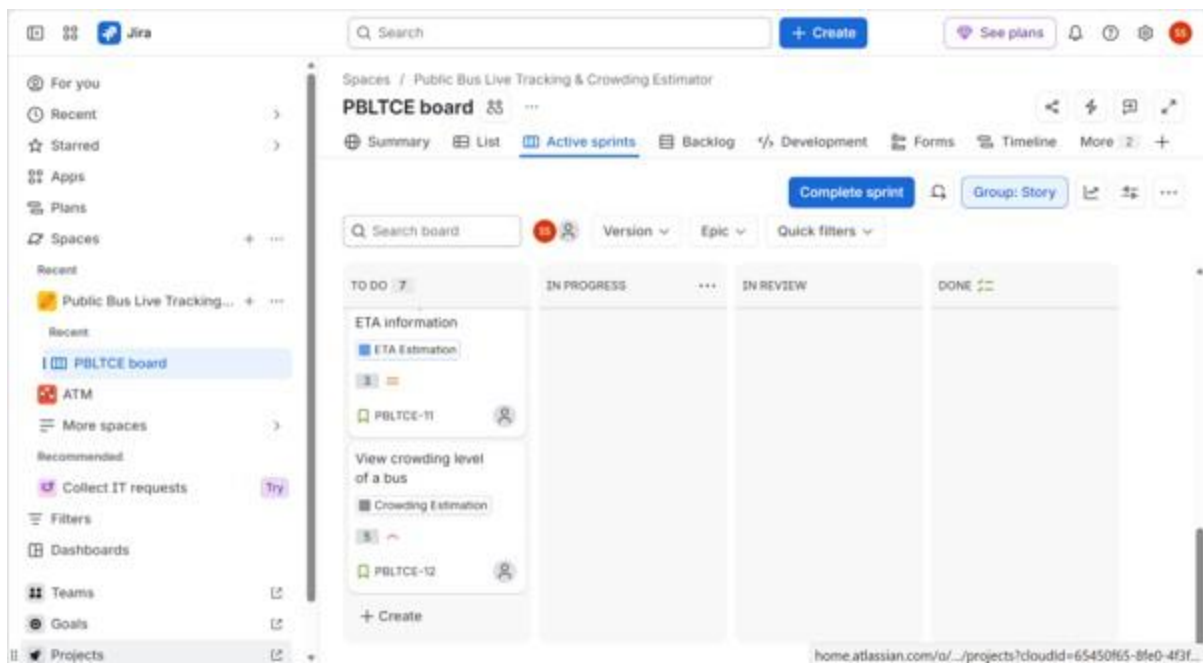
This screenshot shows the Jira Backlog for the 'PBLTCE board' under the space 'Public Bus Live Tracking & Crowding Estimator'. The backlog contains 8 work items, continuing from the previous screenshot. Each item is a user story with a key, description, priority, status, story points, and assignee.

Key	Description	Priority	Status	Story Points	Assignee
PBLTCE-13	Calculate crowding level of buses	Crowding ...	To Do	8	[User]
PBLTCE-14	View updated crowding information	Crowding ...	To Do	3	[User]
PBLTCE-15	Search for available buses	Commute...	To Do	3	[User]
PBLTCE-16	View bus location, ETA, and crowding l...	Commute...	To Do	5	[User]
PBLTCE-17	Display current bus information clearly	Commute...	To Do	3	[User]
PBLTCE-18	View status of all tracked buses	Fleet Mon...	To Do	5	[User]
PBLTCE-19	Identify buses with missing or delayed...	Fleet Mon...	To Do	5	[User]
PBLTCE-20	Monitor crowding information across b...	Fleet Mon...	To Do	3	[User]

Screenshot 2 Sprint 1 – Active Sprint Board

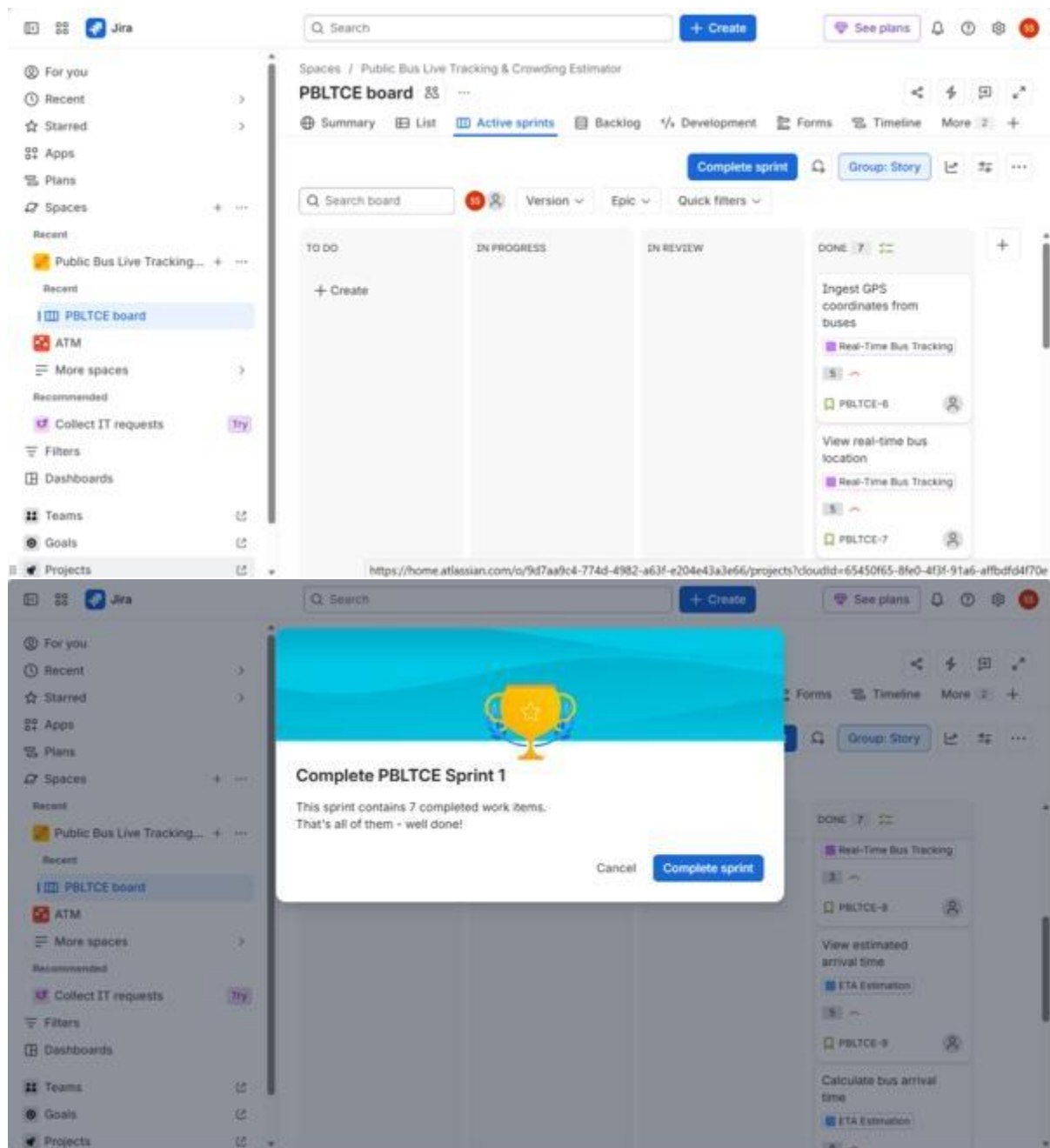


This screenshot shows the Jira interface for the 'Public Bus Live Tracking & Crowding Estimator' space, specifically the 'PBLTCE board' active sprint board. The board is organized into four columns: 'TO DO', 'IN PROGRESS', 'IN REVIEW', and 'DONE'. The 'TO DO' column contains three items: 'Ingest GPS coordinates from buses' (with a sub-item 'Real-Time Bus Tracking'), 'View real-time bus location' (with a sub-item 'Real-Time Bus Tracking'), and 'Monitor GPS data'. The 'IN PROGRESS' column is empty. The 'IN REVIEW' column is empty. The 'DONE' column is empty. The board has a search bar, a 'Complete sprint' button, and a 'Group: Story' button. The left sidebar shows the 'Recent' section with 'Public Bus Live Tracking...' and 'PBLTCE board' listed. The 'Recommended' section shows 'Collect IT requests' with a 'Try' button. The 'Filters' section shows 'Dashboards'. The 'Teams' section shows 'Teams'. The 'Goals' section shows 'Goals'. The 'Projects' section shows 'Projects'.



This screenshot shows the Jira interface for the 'Public Bus Live Tracking & Crowding Estimator' space, specifically the 'PBLTCE board' active sprint board. The board is organized into four columns: 'TO DO', 'IN PROGRESS', 'IN REVIEW', and 'DONE'. The 'TO DO' column contains three items: 'ETA information' (with a sub-item 'ETA Estimation'), 'View crowding level of a bus' (with a sub-item 'Crowding Estimation'), and '+ Create'. The 'IN PROGRESS' column is empty. The 'IN REVIEW' column is empty. The 'DONE' column is empty. The board has a search bar, a 'Complete sprint' button, and a 'Group: Story' button. The left sidebar shows the 'Recent' section with 'Public Bus Live Tracking...' and 'PBLTCE board' listed. The 'Recommended' section shows 'Collect IT requests' with a 'Try' button. The 'Filters' section shows 'Dashboards'. The 'Teams' section shows 'Teams'. The 'Goals' section shows 'Goals'. The 'Projects' section shows 'Projects'.

Screenshot 3 Sprint 1 – Completed Sprint Board



8. SPRINT 2 Sprint Goal

To implement crowding estimation, commuter information, and fleet monitoring functionality.

Sprint 2

- Duration: **1 week**
- Number of User Stories: **8**

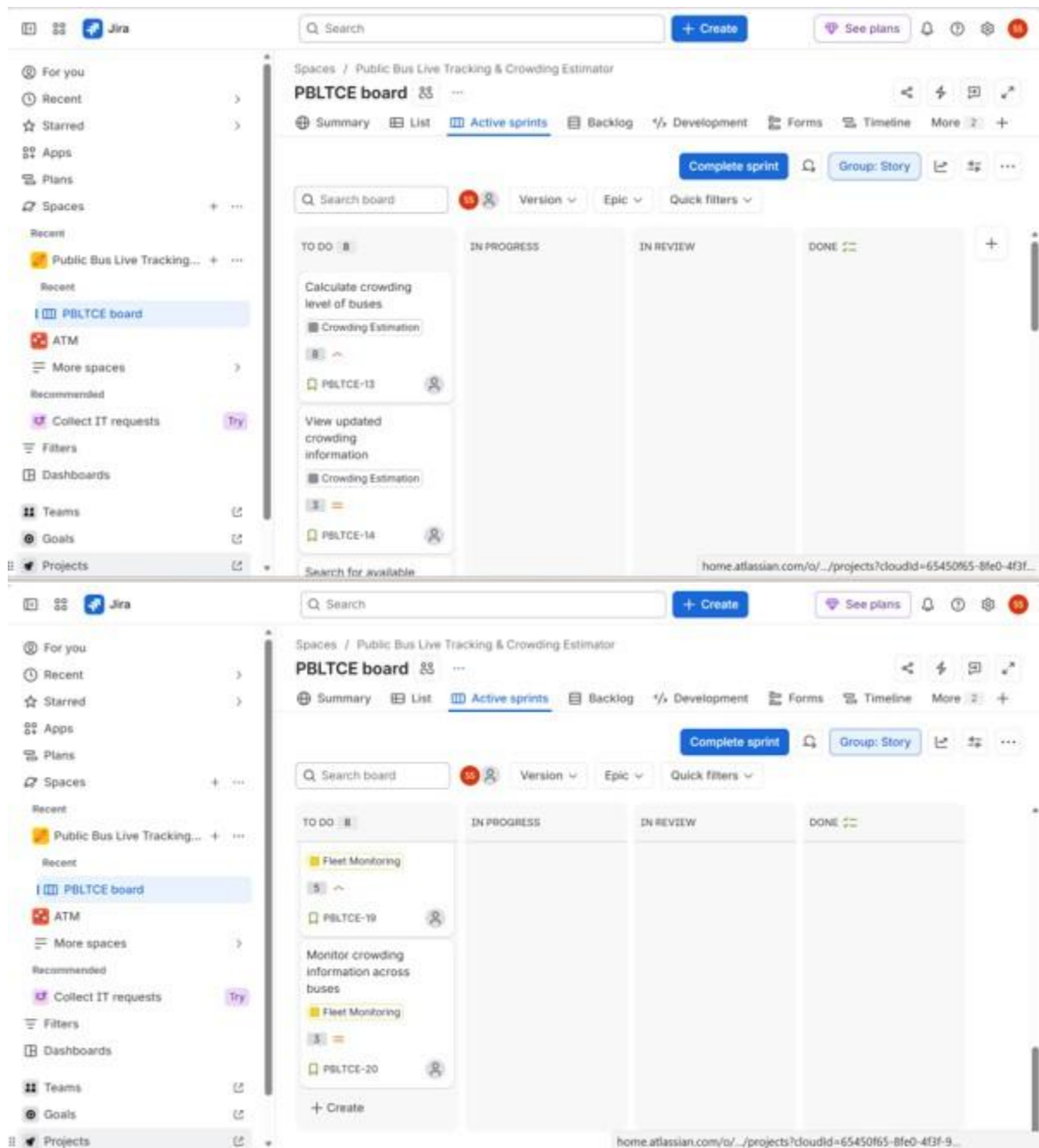
- Total Story Points: **35**

The remaining User Stories were added to Sprint 2 and simulated through:

To Do → In Progress → Done

All eight User Stories were successfully completed and Sprint 2 was completed in Jira.

Screenshot 4 Sprint 2 – Active Sprint Board



Screenshot 5 Sprint 2 – Completed Sprint Board

The screenshot displays a Jira interface for a project named 'Public Bus Live Tracking & Crowding Estimator'. The main view is the 'PBLTCE board', which is currently in the 'Active sprints' tab. The board is organized into four columns: 'TO DO', 'IN PROGRESS', 'IN REVIEW', and 'DONE'. The 'DONE' column contains two tasks, both of which are 'Crowding Estimation' type. The first task is 'Calculate crowding level of buses' (PBLTCE-13) and the second is 'View updated crowding information' (PBLTCE-14). Both tasks are assigned to a user named 'PBLTCE-13'. The board also features a 'Complete sprint' button and a 'Group: Story' button. The left sidebar shows the Jira navigation menu with options like 'For you', 'Recent', 'Starred', 'Apps', 'Plans', 'Spaces', 'Recent', 'Public Bus Live Tracking...', 'PBLTCE board', 'ATM', 'More spaces', 'Recommended', 'Collect IT requests', 'Filters', 'Dashboards', 'Teams', 'Goals', and 'Projects'. The top of the interface includes a search bar, a '+ Create' button, and a 'See plans' button.

Spaces / Public Bus Live Tracking & Crowding Estimator

PBLTCE board

Summary List **Active sprints** Backlog Development Forms Timeline More

Complete sprint Group: Story

Search board Version Epic Quick filters

TO DO IN PROGRESS IN REVIEW DONE

+ Create

Calculate crowding level of buses

Crowding Estimation

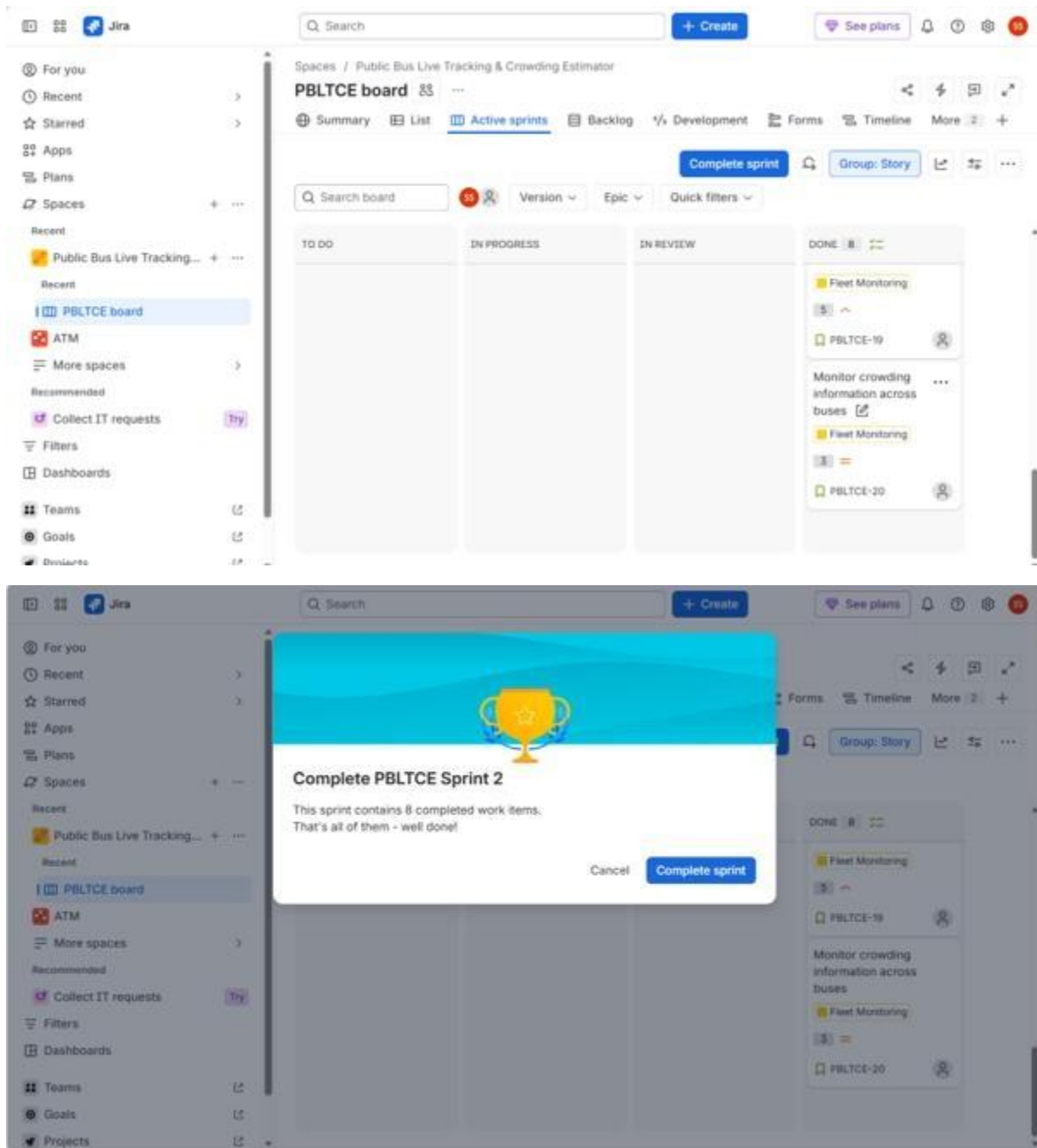
PBLTCE-13

View updated crowding information

Crowding Estimation

PBLTCE-14

Search for available



9. BURNDOWN CHART

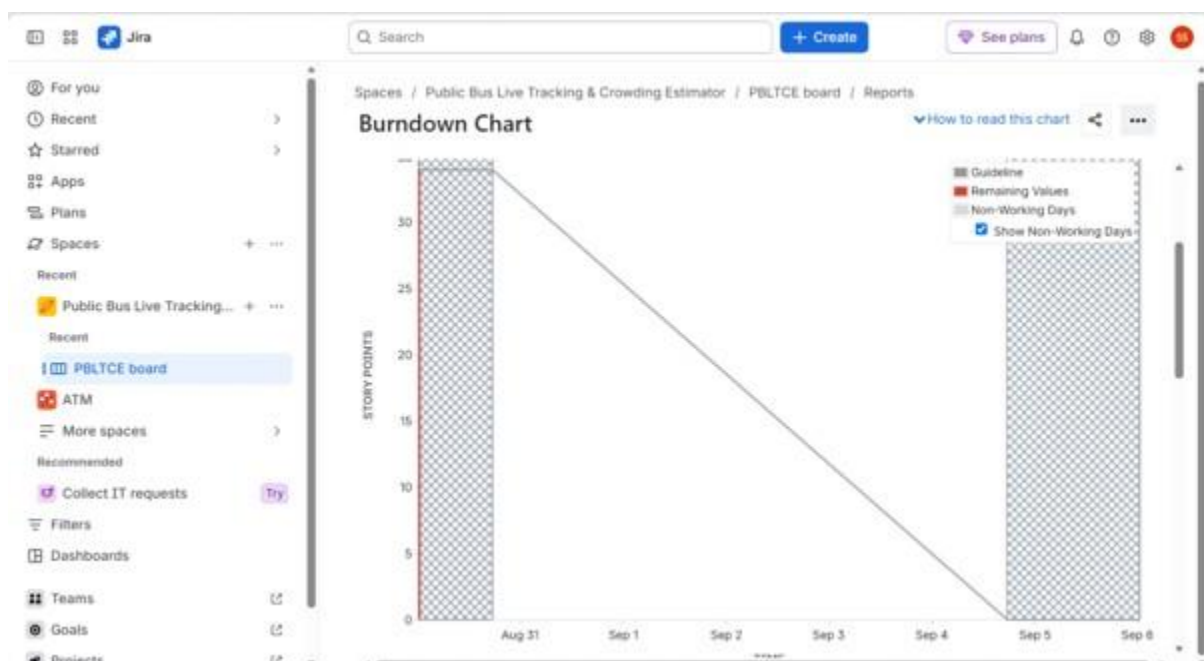
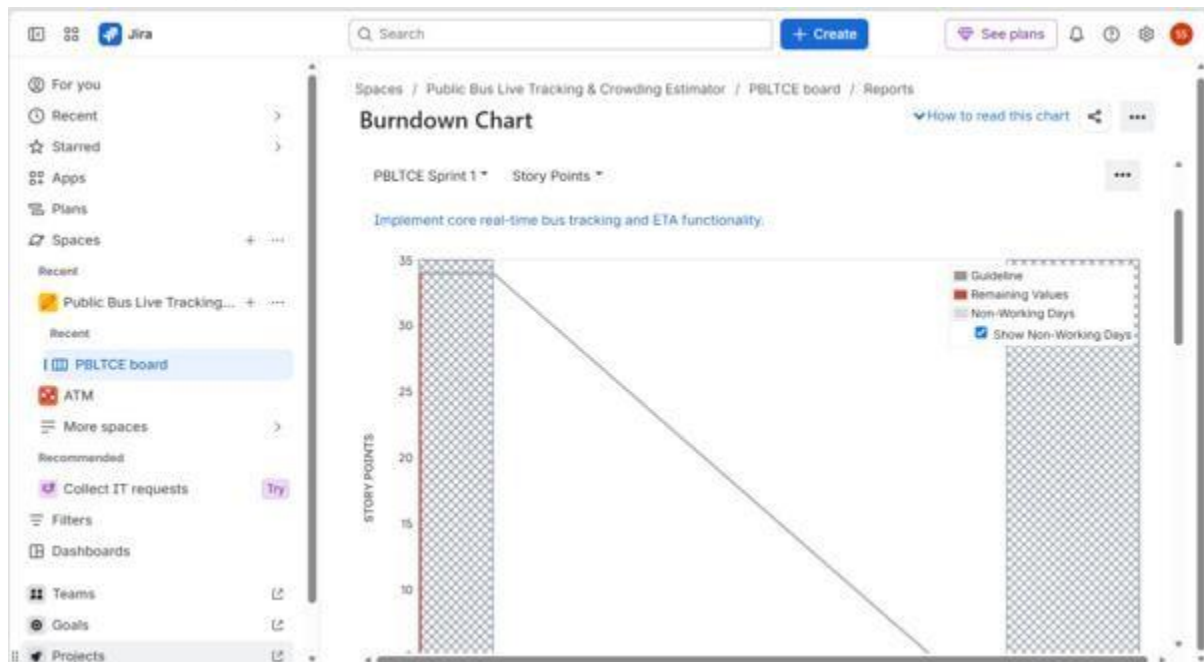
A Burndown Chart was generated in Jira to monitor the amount of work remaining during the Sprints.

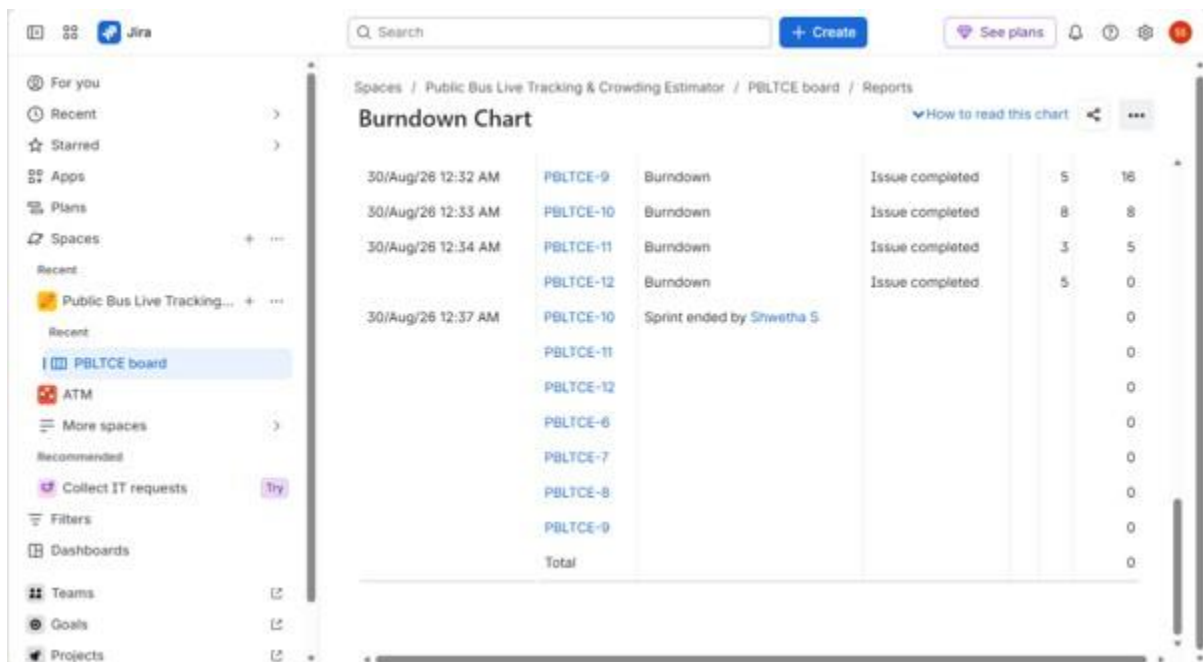
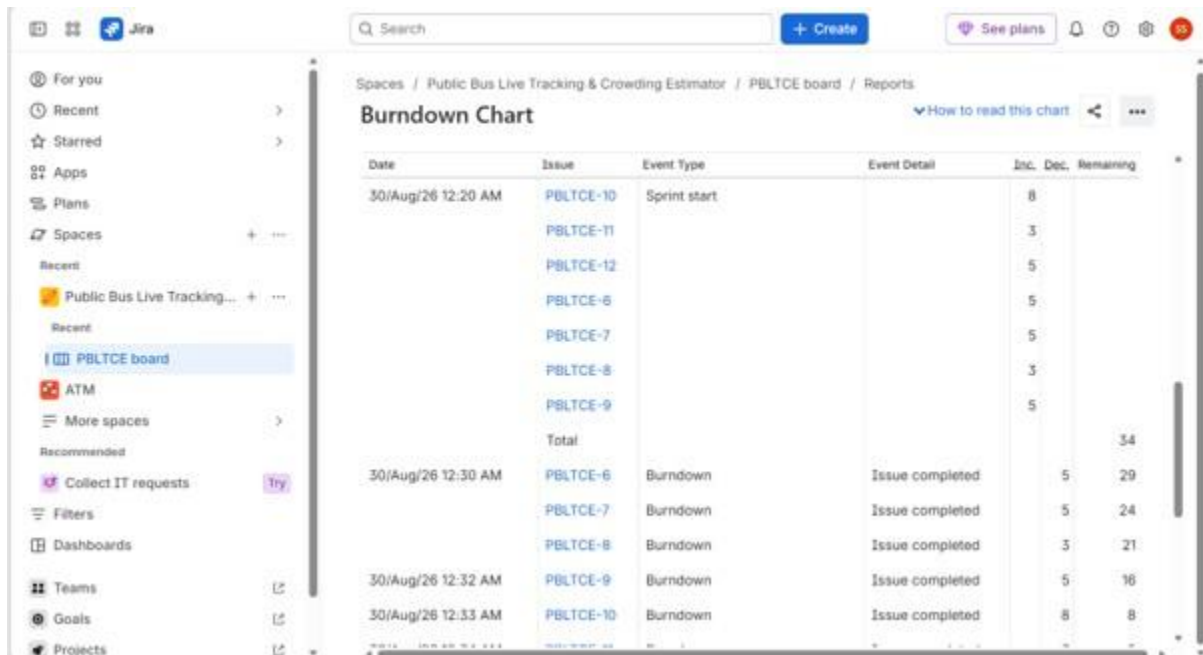
The chart provides a visual representation of the progress of work against the planned Sprint workload.

The Burndown Chart helps the Scrum team:

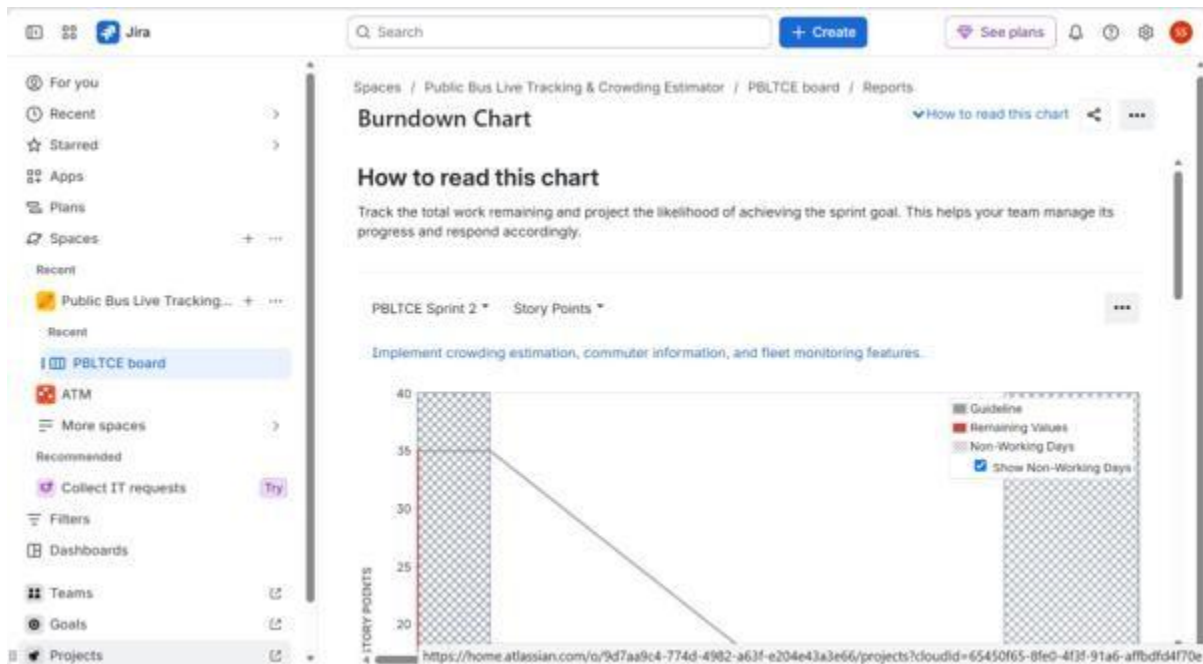
- Monitor remaining work.
- Track the rate at which User Stories are completed.
- Compare actual progress with the expected progress.
- Identify delays or inconsistencies.
- Understand the team's capacity.

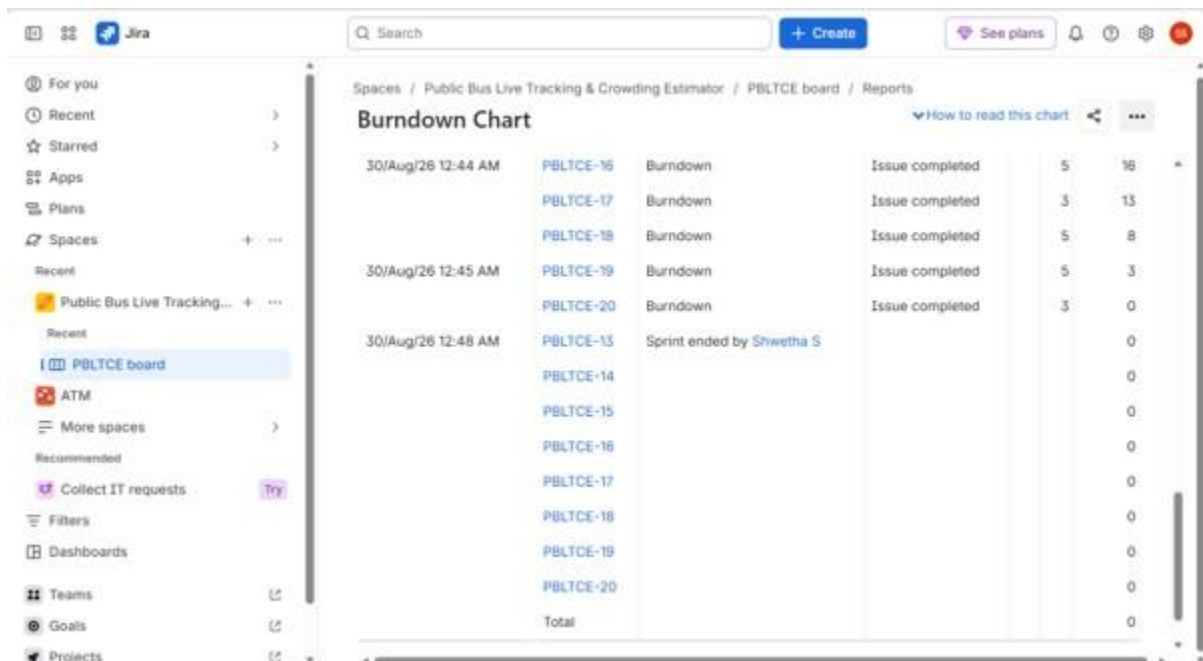
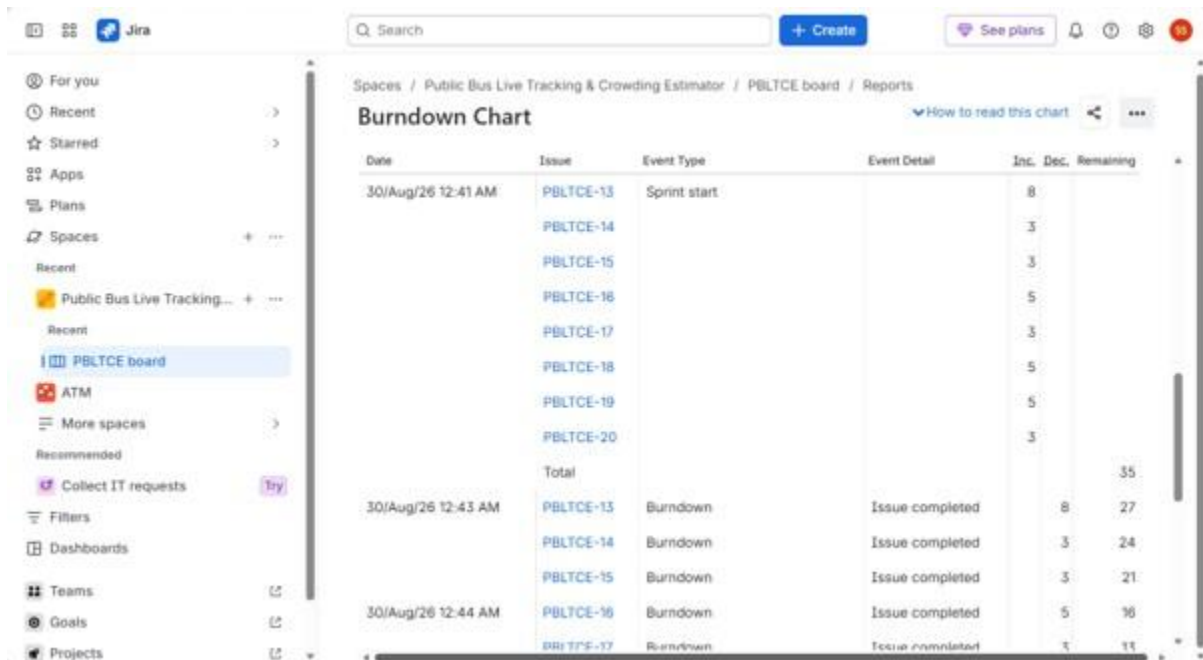
Screenshot 6 Sprint 1 – Burndown Chart





Screenshot 7 Sprint 2 – Burndown Chart





10. REFLECTION

1. Did your estimations reflect the actual effort?

The estimates provided a reasonable relative measure of the effort required for the User Stories. The Fibonacci Story Point system helped differentiate simpler stories from more complex stories.

Since the Sprint was simulated rather than based on actual software development time, the Story Points cannot be directly compared with real implementation hours. However, the estimates were useful for Sprint Planning and distributing the backlog across the two Sprints.

2. Was your backlog well-prioritized?

Yes, the backlog was prioritized according to user impact and the importance of the required functionality.

The higher-priority functionality was considered earlier during Sprint Planning. The User Stories were also organized under their corresponding Epics, which made the backlog structured and easier to manage.

3. How did your simulated Sprint align with your plan?

The simulated Sprints followed the planned Agile Scrum workflow.

Sprint 1 contained seven User Stories and was completed successfully. The remaining eight User Stories were then assigned to Sprint 2.

During both Sprints, the User Stories progressed through:

To Do → In Progress → Done

This simulation demonstrated how work is planned, tracked, and completed within Scrum Sprints.

4. What insights did the Burndown Chart give about your team's capacity?

The Burndown Charts showed how the remaining workload decreased as User Stories were completed.

They provided a visual representation of Sprint progress and helped indicate whether work was being completed according to the planned Sprint workload.

Since this was a simulated exercise, the Burndown Charts mainly demonstrate the relationship between Story Point completion and remaining work. They also provide an indication of how much work the team was able to complete during each Sprint.

11. CONCLUSION

The Jira exercise demonstrated the basic Agile Scrum workflow of converting requirements into Epics and User Stories, prioritizing and estimating the backlog, planning Sprints, tracking work, and analyzing Sprint progress.

Two Sprints were successfully simulated, with all **15 User Stories** completed.

The use of Story Points helped estimate relative effort, while the Burndown Charts provided a visual representation of Sprint progress and remaining workload.

Overall, the exercise provided practical experience in **Agile backlog management, Sprint Planning, Scrum workflow, Story Point estimation, and Sprint progress analysis using Jira.**